

3. What is an R Markdown / Quarto Document?

So far, we have used an R script to write code.

Now we will learn about documents that combine:

- Regular writing
- R code
- Results
- Graphs

These documents are useful for creating reports.

What is a Reproducible Report?

A **reproducible report** combines your explanation and your code in one place.

This is helpful because someone else can see:

1. What question you asked.
2. What code you used.
3. What results you got.
4. How you explained those results.

Tip

A reproducible report helps you avoid copying and pasting results by hand.

Comparing .R, .Rmd, and .qmd Files

File Type	Name	What It Is Used For
.R	R script	Writing and saving R code
.Rmd	R Markdown document	Combining text, R code, and output
.qmd	Quarto document	A newer format for reports, websites, slides, and more

What is R Markdown?

R Markdown is a document format that lets you combine writing, code, and results.

R Markdown files usually end with:

```
.Rmd
```

R Markdown has been widely used in many statistics and data science courses.

What is Quarto?

Quarto is a newer tool for creating documents that include code and results.

Quarto files usually end with:

```
.qmd
```

With Quarto, you can create:

- HTML webpages
- PDF documents
- Word documents
- Slides
- Websites

This guide is written as a Quarto website.

Beginner Note

You do not need to memorize all the differences between R Markdown and Quarto right now.

For this course, focus on this idea:

A Quarto document lets you write explanations, run code, and show results in one file.

Parts of a Rmd /Quarto Document

A simple Quarto document usually has:

1. A title section at the top.
2. Regular text.
3. Code chunks.

Example Title Section

At the top of a Rmd/Quarto file, you may see something like this:

```
---  
title: "My First Rmd Document"  
author: "Your Name"  
format: html  
---
```

This part is called the **YAML header**.

You do not need to write complicated YAML right now. It simply tells Rmd basic information about the document.

Regular Text

You can type normal sentences in a Rmd document.

For example:

```
This report shows my first analysis in R.
```

Code Chunks

A **code chunk** is a special section where R code can be run.

In a real Quarto document, an R code chunk has three parts:

1. A starting line with three backticks followed by `{r}`.
2. One or more lines of R code.
3. An ending line with three backticks.

For example, a code chunk that calculates $2 + 2$ would look like this in your `.Rmd` file:

```
Line 1: three backticks followed by {r}  
Line 2: 2 + 2  
Line 3: three backticks
```

When you render the document, Quarto runs the code and shows the result.

Tip

Code chunks are different from regular text.
Regular text explains your work.
Code chunks run R code.

A Simple Quarto Document

Here is the structure of a small Quarto document:

```
---
title: "My First Quarto Document"
author: "Your Name"
format: html
---

# My First Quarto Document

This is my first Quarto document.

Here is a simple calculation.

Then add an R code chunk that contains:

2 + 2

After rendering, the answer appears below the code.
```

In your actual `.qmd` file, the R code chunk would start with three backticks followed by `{r}`, and end with three backticks.

What is Rendering?

To **render** a document means to turn it into a finished report.

For example, In Quarto and R Markdown, `.qmd` or `.Rmd` file can turn into:

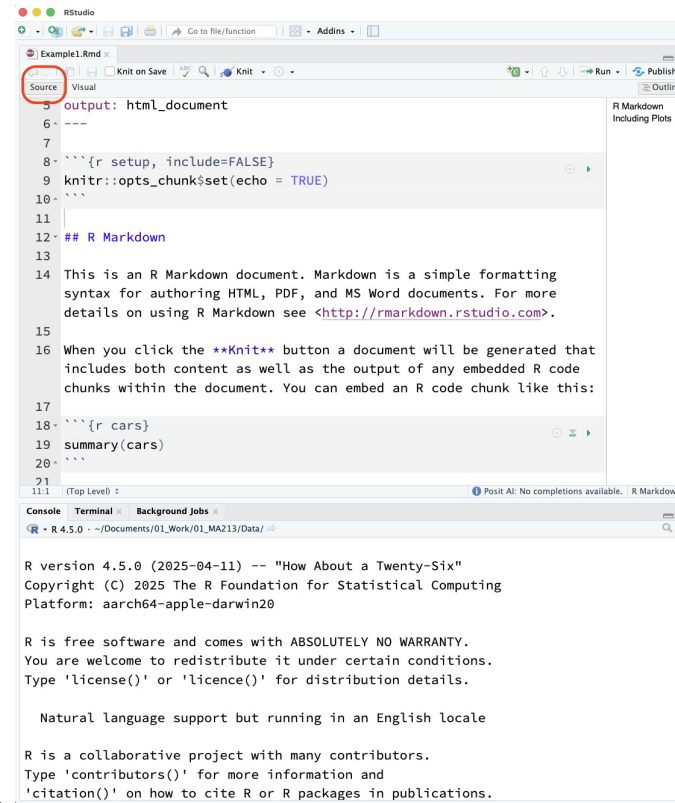
- an HTML webpage,
- a PDF document,
- or a Word document.

In RStudio, you can usually click the **Render** button.

Interface in Rmd/Quarto Documents

When you open a .Rmd or .qmd file in RStudio, you will see:

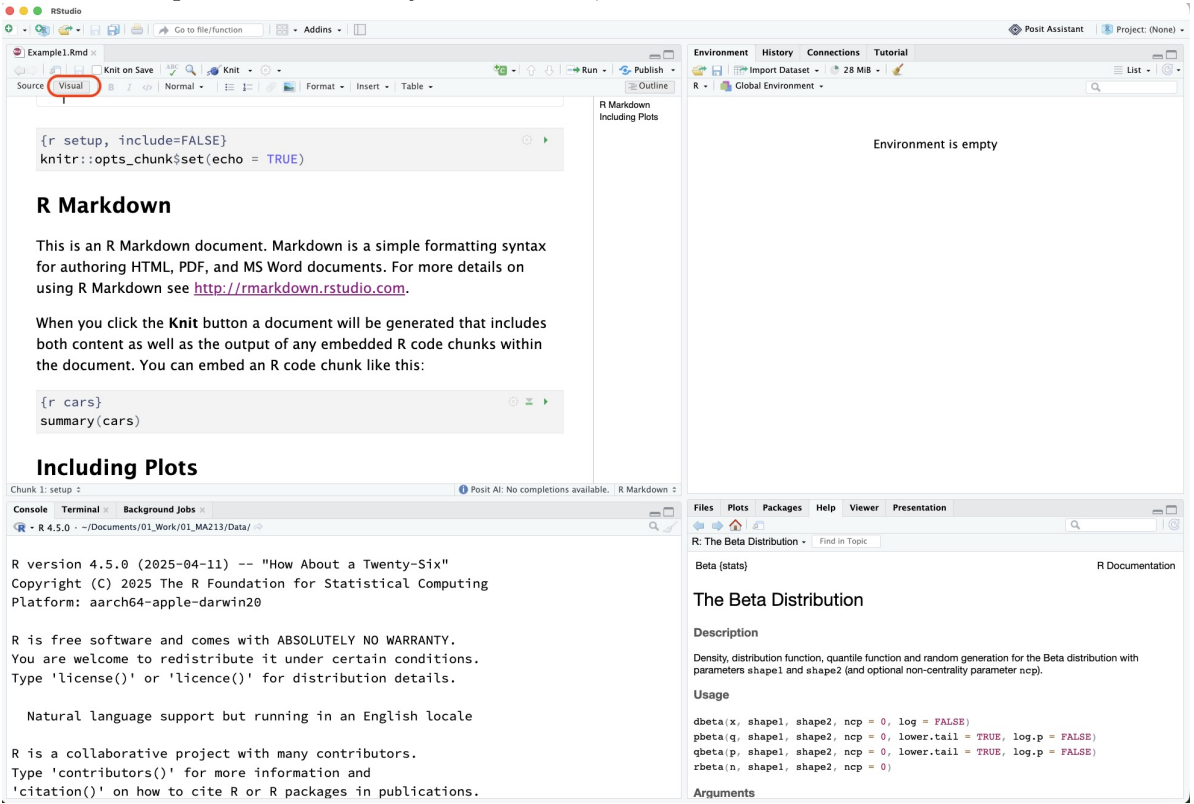
Source View



Click **Source** to see the raw text and code in your document.

Visual View

Click **Visual** to see a more polished version of your document, with formatted text and rendered



The screenshot displays the RStudio interface with the 'Visual' view selected, indicated by a red circle around the 'Visual' button in the top-left toolbar. The main editor window shows a rendered R Markdown document. It includes an R code chunk at the top with the following code:

```
{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

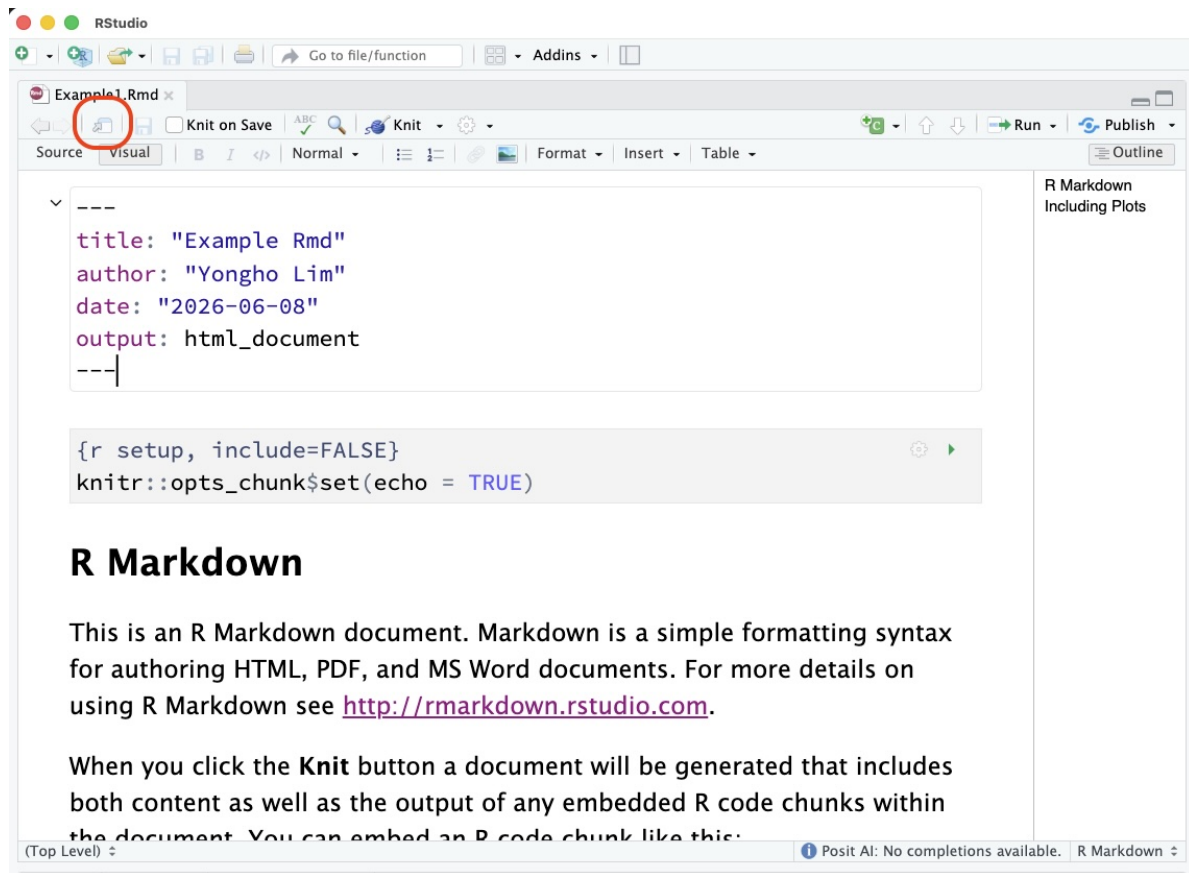
Below the code chunk, the document content is rendered with formatted text. It starts with the heading 'R Markdown', followed by a paragraph explaining that this is an R Markdown document and providing a link to <http://rmarkdown.rstudio.com>. Another paragraph explains that clicking the Knit button generates a document including both content and R code chunk outputs. This is followed by another R code chunk:

```
{r cars}
summary(cars)
```

Below this, the heading 'Including Plots' is visible. The bottom-left pane shows the console output for 'Chunk 1: setup', displaying R version information and copyright details. The bottom-right pane shows the 'R: The Beta Distribution' documentation page, which includes a description of the Beta distribution and its usage, with example R code snippets like `dbeta(x, shape1, shape2, ncp = 0, log = FALSE)`.

code chunks.

Show in a new window



Click **Show in new window** to see the rendered document in a separate window.

